

SAIRAM SRIDHARAN

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EDUCATION

Northeastern University | Boston, MA

Expected: May 2026

Masters of Robotics, Electrical and Computer Engineering

Related Coursework: Robot Sensing and Navigation, Mobile Robotics, Robot Mechanics and Control, Field Robotics, Assistive Robotics, Reinforcement Learning, Optimization Algorithms, Non-Linear Controls.

Vellore Institute of Technology | Chennai

May 2024

Bachelor of Technology, Mechatronics and Automation

Related Coursework: Industrial Automation, PLC, Machine Learning, Computer Vision, IoT, Robot Mechanics, Product Design and Development.

TECHNICAL SKILLS

Programming: Python, C++, Rust

Software and Tools: ROS, MATLAB, SIMULINK, Open CV, MuJoCo, PyTorch, Arduino IDE, Git, Docker SolidWorks

WORK EXPERIENCE

Research Intern | WIKA | Chennai

June 2022 - July 2022

- Assisted the team in design and production of Industrial sensors, including flow, temperature, and pressure measurement devices
- Contributed to calibrating Industrial sensors to maintain the accuracy and precision of pressure and temperature measurement instruments.

Research Intern | Lucas TVS | Chennai

May 2022 - June 2022

- Monitored and supervised assembly lines for alternators, starters, motors, and wipers used in automotive applications, gaining practical experience in production workflows and process optimization.
- Worked with the R&D department to research product innovations and evaluate emerging technologies for future development.

ACADEMIC PROJECTS

Autonomous Jenga-Playing Robot | Northeastern University | [Robotic Manipulation – Jenga](#)

- Built a robotic system using the WidowX-250 arm to autonomously play Jenga by integrating perception, control, and mechanical design for real-world manipulation tasks.
- Implemented AprilTag-based block detection, real-time pose transformation, and closed-loop feedback control in ROS 2, achieving successful block extraction without tower collapse.

Camera–LiDAR Extrinsic Calibration On-the-fly | Northeastern University | [Online Camera–LiDAR Calibration](#)

- Developed an online camera–LiDAR extrinsic calibration pipeline using edge-based alignment of projected 3D LiDAR features with 2D image edges.
- Validated robustness through synthetic offset injection and recovery experiments, demonstrating suitability for real-world robotic deployment.

Structure-from-Motion (SfM) 3D Reconstruction Pipeline | Northeastern University | [SfM 3D Reconstruction](#)

- Built a complete SfM pipeline from scratch, including feature extraction, epipolar geometry, pose recovery, triangulation, PnP and global bundle adjustment using GTSAM.
- Validated reconstruction accuracy by comparing outputs with COLMAP and extracting real camera intrinsics (K) and radial distortion (k_1) from COLMAP calibration.

Adaptive Path-Guided RRT (APG-RRT) Implementation | Northeastern University | [APG-RRT Path Planning](#)

- Implemented and evaluated APG-RRT as an efficient alternative to RRT, generating smoother, shorter, and collision-free trajectories in structured environments (corridors, bends, narrow passages).
- Extended the algorithm from 2D Python simulations into a ROS 2 framework, integrating with TurtleBot3 in Gazebo to validate autonomous navigation in realistic robotic scenarios.

ORB-SLAM3 on Boston Dynamics' Spot Robot | Northeastern University

- Implemented ORB-SLAM3 on data collected from the camera on the Boston Dynamics Spot robot.
- Assessed the accuracy and robustness of monocular SLAM in dynamic environments.

Route Master: Navigating Dynamic Environments with Advanced Mapping | VIT University | [Autonomous Mobile Robot](#)

- Developed a cost-effective autonomous mobile robot using LiDAR and Raspberry Pi, integrated with ROS for both tele-operated and autonomous navigation.
- Performed iterative testing to achieve reliable performance in dynamic indoor environments, with the robot effectively avoiding obstacles and following predefined paths with high precision.